When Does Adaptive Alpha Control Help? A Controlled Comparison of Adversarial Debiasing and Post-Processing Calibration

Lewis Shearer

October 26, 2026

Abstract

Adversarial debiasing forces a single hand-picked weight (α) to govern the accuracy-fairness trade-off for an entire training run. We propose Adaptive Alpha Control, a dual-ascent controller that adjusts two separate weights – one for an equal-opportunity constraint, one for an average-odds constraint – every epoch, with each target set relative to the task's own measured baseline bias rather than an arbitrary constant. On COMPAS (30 seeds, paired significance testing), the controller reaches a different point on the accuracy-fairness frontier automatically: statistically indistinguishable accuracy from the best fixed alpha found by sweeping, significantly better equal-opportunity gap ($p < 0.0001$), but significantly worse average-odds gap ($p = 0.027$) – a trade, not a dominance. We then compare both against post-processing equalized-odds calibration (Hardt et al., 2016), which requires no adversary or retraining. On COMPAS it substantially outperforms every adversarial variant tested, including ours, on every metric simultaneously. Testing whether this generalizes, we find it does not: on a smaller second dataset (German Credit, 200 test rows vs. COMPAS's 1,056), post-processing becomes the *worst* method, significantly so on the average-odds gap, while every adversarial method – including ours – stays stable across both datasets. The paper's contribution is therefore not a ranking of methods but a controlled comparison identifying the condition (available calibration-set size) under which each approach should be preferred.

1 Introduction

Adversarial debiasing [Zhang et al., 2018] trains a predictor against an adversary that tries to recover a sensitive attribute from the predictor's output, scaling how much the predictor listens to the adversary by a single weight, α. α is fixed and chosen by hand before training begins – but the right value is not constant across a training run: early on, an undertrained predictor's output may carry little signal for the adversary to exploit regardless of α; later, as the predictor converges, bias can re-emerge in ways a value chosen at the start cannot anticipate. A single static α is therefore a compromise averaged over the whole run rather than a value fit to what is actually happening at each point in training, and finding even a reasonable fixed value requires sweeping several candidates and comparing their outcomes after the fact.

This paper replaces the fixed weight with a controller that adjusts two separate weights – one for each of two fairness criteria (equal opportunity and average odds) – every epoch, based on how far the current model sits from a fairness target set relative to the task's own measured baseline bias. The controller is framed as dual ascent on a constrained optimization problem rather than a hand-built heuristic, giving each design choice (what the update does, why two separate weights

rather than one, why a safety floor on accuracy) a principled justification rather than a post-hoc one.

Having built this, the natural next question is whether a fairness mechanism that intervenes during training is actually necessary at all. We compare against post-processing equalized-odds calibration [Hardt et al., 2016], which requires no adversary and no retraining – only a calibration pass on an already-trained, otherwise-unconstrained predictor. On our primary dataset (COMPAS), post-processing substantially outperforms every adversarial variant tested, including our own adaptive mechanism. Rather than treat this as a reason to abandon the adversarial approach, we test whether the result generalizes to a second, smaller dataset (German Credit) and find that it does not: post-processing's advantage specifically depends on having enough held-out data to calibrate reliable per-group thresholds, a condition COMPAS meets and German Credit does not. The paper's contribution is therefore not "method X beats method Y," but a controlled comparison that identifies the condition under which each approach is preferable.

Contributions:

1. Adaptive Alpha Control: a dual-ascent controller for the predictor/adversary weight in adversarial debiasing, with two separate multipliers for equal opportunity (DEO) and average-odds (DAO) gaps, each targeting a fraction of the task's own measured baseline bias rather than an arbitrary fixed constant.

2. A rigorous comparison against a fixed-alpha sweep baseline (3 points: $\alpha \in \{0.5, 1.0, 2.0\}$) across 30 seeds, showing the adaptive method lands on or near the best fixed-alpha operating point without requiring the sweep.

3. A controlled comparison against post-processing equalized-odds calibration [Hardt et al., 2016], showing it dominates every adversarial method on COMPAS but becomes specifically unreliable on the compound fairness metric (DAO) when the calibration set is small (German Credit), a conditional rather than universal result.

4. Cross-dataset validation (COMPAS, German Credit) with paired significance testing (Wilcoxon signed-rank, Holm-corrected).

2 Related Work

Adversarial debiasing. Zhang et al. [2018] introduce a predictor/adversary minimax game where a predictor is trained to accomplish a task while an adversary tries to recover a sensitive attribute from only the predictor's output. The predictor's loss subtracts the adversary's (scaled by a weight α), so the predictor is pushed to make its output uninformative about the sensitive attribute to the extent α demands. α is fixed and hand-chosen in the original formulation; we replace it with a per-epoch controller.

Constrained optimization for fairness. Treating a fairness gap as a constraint rather than a soft penalty term has a history in Agarwal et al. [2018] and Cotter et al. [2019], who use Lagrangian-style dual methods to directly target a fairness constraint via alternating gradient ascent on a multiplier. Our dual-ascent controller is the same family of idea applied inside a per-epoch training loop rather than as an outer reduction over a black-box classifier.

Post-processing. Hardt et al. [2016] show that, given an already-trained (unconstrained) classifier's scores, per-group decision thresholds can be chosen – via randomized mixing between two thresholds per group – to exactly satisfy equalized odds while maximizing accuracy, by searching the intersection of each group's ROC convex hull. This requires no retraining and no adversary, but does require enough held-out data per group to estimate a reliable ROC curve.

Domain-adversarial training. Ganin & Lempitsky [2015] introduce the gradient-reversal layer, letting a minimax game of this kind be trained with a single joint backward pass instead of alternating updates. We use this as an implementation-level ablation (Section 6) to check whether our alternating-update implementation itself was costing anything relative to the more standard DANN-style setup – it was not; the two are equivalent.

COMPAS and algorithmic recidivism prediction. Angwin et al. [2016] is the ProPublica investigation that established COMPAS's racial disparity in false positive rates as a motivating real-world case for equalized-odds-style fairness criteria, and is the dataset used throughout this paper's primary experiments.

3 Method

3.1 Predictor/adversary architecture

Predictor: Dense(32, relu) → Dense(16, relu) → Dense(1, sigmoid). Input is the full feature vector; output is a single predicted probability for the task label y.

Adversary: Dense(16, relu) -¿ Dense(8, relu) -¿ Dense(1, sigmoid). Input is the predictor's single scalar output $\hat{y}$ – not the raw features, not the true label – so the adversary can only succeed if $\hat{y}$ itself leaks the sensitive attribute z.

3.2 Fairness metrics

Let TPR_g and FPR_g denote the true/false positive rate within sensitive-attribute group $g \in \{0, 1\}$.

$$\mathrm{DEO} = |\mathrm{TPR}_0 - \mathrm{TPR}_1| \tag{1}$$

$$\mathrm{DAO} = \tfrac{1}{2}\left(|\mathrm{TPR}_0 - \mathrm{TPR}_1| + |\mathrm{FPR}_0 - \mathrm{FPR}_1|\right) \tag{2}$$

Both are computed on held-out data; 0 is perfectly fair by this definition, higher is more biased.

3.3 Baseline adversarial debiasing (fixed alpha)

Following Zhang et al. [2018], for a fixed scalar α:

$$L_P = \mathrm{BCE}(y, \hat{y}) - \alpha \cdot \mathrm{BCE}(z, \hat{z}) \tag{3}$$

$$L_A = \mathrm{BCE}(z, \hat{z}) \tag{4}$$

Each minibatch: update the adversary first (predictor's weights held fixed for that update), then update the predictor (adversary's weights held fixed). This alternation is necessary because L_P's second term depends on the adversary's current best response – a stale or simultaneously-updating adversary would make the "maximize the adversary's loss" signal incorrect.

3.4 Adaptive Alpha Control (dual ascent)

Reframe the problem as constrained optimization:

$$\min \quad \mathrm{BCE}(y, \hat{y}) \tag{5}$$

$$\text{s.t.} \quad \mathrm{DEO} \leq \epsilon_{\mathrm{DEO}}, \quad \mathrm{DAO} \leq \epsilon_{\mathrm{DAO}} \tag{6}$$

with two separate dual variables $\alpha_{\mathrm{DEO}}, \alpha_{\mathrm{DAO}} \geq 0$ updated by gradient ascent on the Lagrangian after every epoch:

$$\alpha_{\mathrm{DEO}} \leftarrow \max(0,\ \alpha_{\mathrm{DEO}} + \eta \cdot (\mathrm{DEO} - \epsilon_{\mathrm{DEO}})) \tag{7}$$

$$\alpha_{\mathrm{DAO}} \leftarrow \max(0,\ \alpha_{\mathrm{DAO}} + \eta \cdot (\mathrm{DAO} - \epsilon_{\mathrm{DAO}})) \tag{8}$$

To make the two multipliers genuinely independent rather than redundant (a single shared adversary scaled by two numbers that would just add together), the adversary is conditioned on the true label: two separate adversary heads are trained, one on the subset of each batch where $y = 1$ (a direct proxy for TPR-gap / DEO leakage) and one where $y = 0$ (adds the FPR-gap component DAO needs):

$$L_P = \mathrm{BCE}(y, \hat{y}) - \alpha_{\mathrm{DEO}} \cdot L_{A,y=1} - \alpha_{\mathrm{DAO}} \cdot \tfrac{1}{2}(L_{A,y=1} + L_{A,y=0}) \tag{9}$$

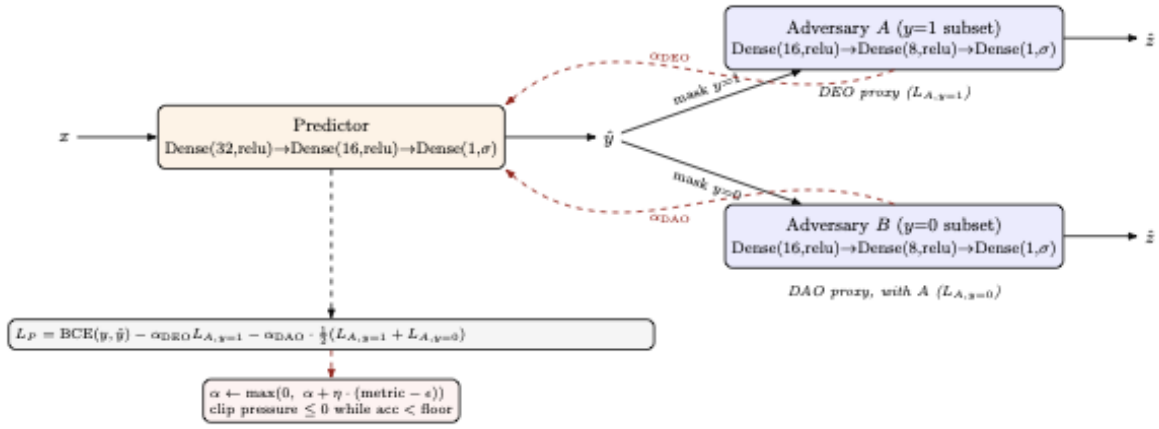


Figure 1: Adaptive Alpha Control: the predictor's output is split by true label into two subsets, each feeding a separate adversary head. Each head's loss proxies one fairness metric's leakage (DEO for the $y = 1$ subset, DAO for both); dual ascent on $\alpha_{\mathrm{DEO}}, \alpha_{\mathrm{DAO}}$ feeds back into the predictor's loss every epoch.

$\epsilon_{\mathrm{DEO}}, \epsilon_{\mathrm{DAO}}$ are set as a fraction (30%) of the dataset's own measured no-mitigation baseline DEO/DAO – not an arbitrary absolute constant – so the target is meaningful regardless of how biased the dataset is to begin with. A hard safety clip prevents the ascent step from ever increasing alpha while accuracy is below a floor (set from the dataset's own measured baseline accuracy): if accuracy is below the floor, the pressure term is clamped to be non-positive that epoch, so alpha can still fall but never climb while accuracy is compromised.

3.5 Post-processing equalized-odds calibration

Following Hardt et al. [2016]: train a plain unconstrained predictor (no adversary, full task accuracy). On held-out calibration data, compute each group's ROC curve and its upper concave

hull (the set of (FPR, TPR) points achievable by randomly mixing two thresholds). Search for the (FPR^*, TPR^*) point achievable by *both* groups' hulls simultaneously that maximizes overall accuracy; realize it per group via randomized mixing between the two empirical thresholds bracketing that point on the group's hull. Evaluated on a disjoint held-out split from the one used for calibration, to avoid optimistic bias.

4 Experimental Setup

4.1 Datasets

COMPAS [Angwin et al., 2016]: 7,214 rows, filtered following the original ProPublica analysis (screening within 30 days of arrest, valid recidivism data, non-traffic charges, a produced COMPAS score), restricted to the two groups ProPublica's own comparison focuses on (African-American, Caucasian) for a clean binary sensitive attribute. Target: two-year recidivism. After filtering: 5,278 rows (4,222 train / 1,056 test), 12 features.

German Credit: 1,000 rows, no filtering needed (800 train / 200 test, 56 features after one-hot encoding). Target: good vs. bad credit risk. Sensitive attribute: gender (binary in this dataset). Included specifically because it is much smaller than COMPAS (200 test rows vs. COMPAS's 1,056), which is exactly the condition the post-processing comparison turns out to depend on.

4.2 Conditions compared

Four conditions, identical predictor architecture, preprocessing, and 30-seed protocol across all of them so the only difference between conditions is the debiasing mechanism itself:

1. **Baseline**: predictor only, no adversary, no debiasing attempt – the reference point for "what does bias look like unmitigated."
2. **Fixed alpha**: Zhang et al.'s adversarial game at $\alpha \in \{0.5, 1.0, 2.0\}$.
3. **Adaptive** (ours): dual-ascent two-constraint controller, Section 3.4.
4. **Post-processing**: Hardt et al.'s equalized-odds calibration, Section 3.5.

4.3 Statistical methodology

30 seeds per condition, paired by seed. The train/test split itself is fixed (identical across every seed and condition); only weight initialization and minibatch shuffling vary with the seed. This isolates the comparison to training stochasticity alone and makes the paired test exact rather than approximate – condition A, seed s and condition B, seed s differ only in the debiasing mechanism, not in what data they saw. Significance assessed via the Wilcoxon signed-rank test on paired per-seed values, with the Holm step-down procedure correcting for the multiple comparisons made across conditions and metrics (src/significance_test.py).

5 Results

5.1 COMPAS

Paired Wilcoxon signed-rank tests, Holm-corrected across all 12 comparisons (full table in `results/significance_`

Condition	Accuracy	DEO	DAO
Baseline (no mitigation)	0.657 ± 0.005	0.249 ± 0.012	0.207 ± 0.015
Fixed $\alpha = 0.5$	0.648 ± 0.005	0.244 ± 0.014	0.203 ± 0.020
Fixed $\alpha = 1.0$	0.638 ± 0.010	0.227 ± 0.022	0.192 ± 0.019
Fixed $\alpha = 2.0$	0.615 ± 0.011	0.167 ± 0.023	0.131 ± 0.023
Adaptive (ours)	0.609 ± 0.009	0.127 ± 0.019	0.144 ± 0.014
Post-processing	0.631 ± 0.020	0.061 ± 0.045	0.060 ± 0.040

Table 1: COMPAS, 30 seeds, mean ± std.

- **Adaptive vs. best fixed alpha** ($\alpha = 2.0$): accuracy not significantly different ($p_{\text{holm}} = 0.087$); DEO significantly better for adaptive ($p_{\text{holm}} < 0.0001$); DAO significantly worse for adaptive ($p_{\text{holm}} = 0.027$). Adaptive reaches a different point on the frontier automatically, not a strictly dominant one.
- **Post-processing vs. adaptive**: significantly better on all three metrics ($p_{\text{holm}} < 0.001$ for each).
- **Adaptive vs. baseline**: a real, significant accuracy cost ($p_{\text{holm}} = 1.2 \times 10^{-5}$) buys significant DEO and DAO reduction (both $p_{\text{holm}} = 2.2 \times 10^{-8}$).
- **Post-processing vs. baseline**: smaller significant accuracy cost than adaptive's, with larger significant DEO/DAO reduction (all $p_{\text{holm}} < 3 \times 10^{-5}$).

Figure 2 shows accuracy against DEO and DAO across every condition; post-processing sits clearly above and to the right of every adversarial variant, including adaptive.

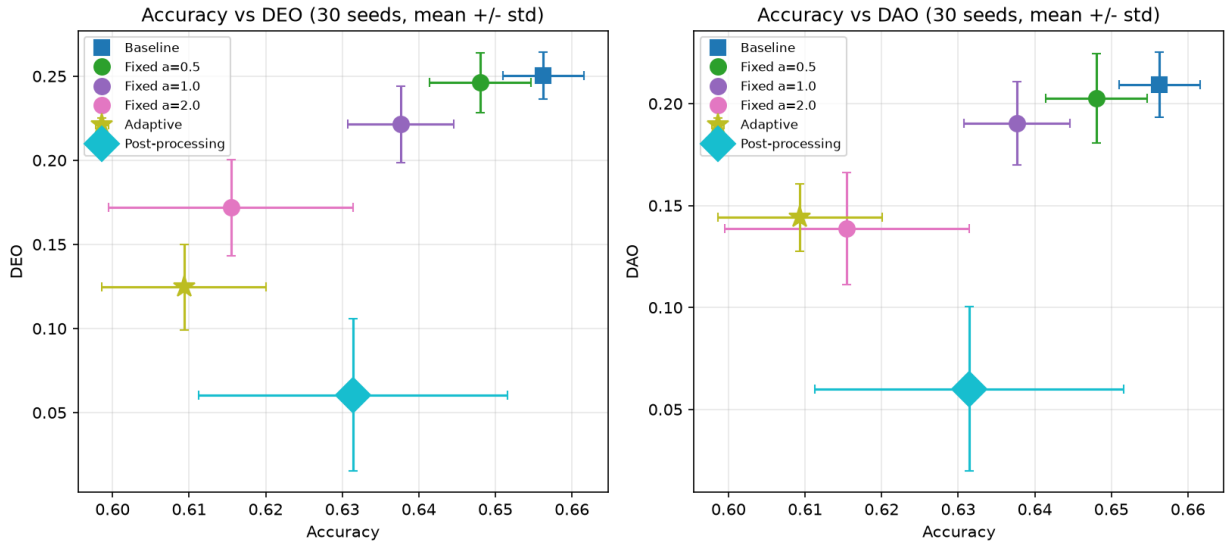


Figure 2: Accuracy vs. DEO/DAO, COMPAS, 30 seeds (mean ± std). Post-processing (diamond) dominates every adversarial variant.

Condition	Accuracy	DEO	DAO
Baseline (no mitigation)	0.721 ± 0.016	0.058 ± 0.037	0.053 ± 0.025
Fixed $\alpha = 0.5$	0.726 ± 0.016	0.063 ± 0.039	0.056 ± 0.030
Fixed $\alpha = 1.0$	0.729 ± 0.015	0.045 ± 0.027	0.057 ± 0.027
Fixed $\alpha = 2.0$	0.726 ± 0.015	0.047 ± 0.028	0.053 ± 0.023
Adaptive (ours)	0.722 ± 0.016	0.049 ± 0.036	0.049 ± 0.025
Post-processing	0.711 ± 0.036	0.061 ± 0.055	0.105 ± 0.067

Table 2: German Credit, 30 seeds, mean ± std.

5.2 German Credit

The pattern reverses: post-processing is the *worst* condition on accuracy and DAO, with roughly double the variance of every other condition on every metric. Paired testing (full table in `results/significance_ge` Holm-corrected across 15 comparisons) confirms this is real for DAO specifically: post-processing vs. adaptive, vs. fixed $\alpha = 2.0$, and vs. baseline are all significant ($p_{\text{holm}} = 0.0012, 0.0011, 0.0030$ respectively). DEO and accuracy differences are directionally the same but not statistically significant at $n = 30$ on this smaller dataset. Adaptive vs. baseline shows no significant difference on any metric on German – consistent with German's low baseline bias leaving little for any debiasing mechanism to fix.

6 Discussion

Adaptive alpha control's actual claim. On COMPAS, paired significance testing (Wilcoxon, Holm-corrected) shows the dual-ascent controller is not simply "comparable" to the best fixed alpha found by sweeping – it is a genuine trade differently shaped. Against fixed $\alpha = 2.0$: accuracy is not significantly different (0.609 vs. 0.615, $p_{\text{holm}} = 0.087$), DEO is significantly *better* (0.127 vs. 0.167, $p_{\text{holm}} < 0.0001$), and DAO is significantly *worse* (0.144 vs. 0.131, $p_{\text{holm}} = 0.027$). The honest claim is therefore "the controller finds a different, not strictly better, point on the frontier automatically" – it over-weights the DEO constraint relative to DAO at this operating point, which is a property of the two dual-ascent multipliers' relative convergence rates, not an error. A practitioner who cares specifically about equal opportunity (DEO) gains from the adaptive mechanism without searching; one who weights DAO more would still need to sweep.

Post-processing wins on COMPAS, but the win is conditional. The natural follow-up question to "post-processing beats adversarial training" is *why would anyone use adversarial training at all, then* – and the German Credit result answers it: post-processing's advantage depended on COMPAS's comparatively large held-out set. Adversarial methods (fixed-alpha and adaptive) use the full training set during optimization and do not share this weakness; they stayed stable across both datasets while post-processing's DAO specifically degraded, significantly, on the smaller one. The practical guidance this supports: check how much held-out data is actually available for calibration before choosing a debiasing approach based on a result like ours on COMPAS alone.

Why German moved so little. German Credit's own baseline DEO/DAO (0.058 / 0.053) are already small relative to COMPAS's (0.250 / 0.209) – gender disparity in this dataset is simply a smaller problem than racial disparity in COMPAS's recidivism predictions to begin with. Every adversarial method tested landed within noise of baseline on German, which is an honest finding about the dataset, not a failure of the method: there was comparatively little bias to remove.

Ablations confirm rather than improve the mechanism. Two alternative designs were

tested against the dual-ascent controller: annealing $\epsilon_{\text{DEO}}, \epsilon_{\text{DAO}}$ from the baseline gap down to their final targets (rather than fixing them from epoch 1), and replacing the alternating-update training procedure with a single joint gradient-reversal backward pass. Annealing traded DEO reduction for accuracy – a different point on the same frontier, not a better one. Gradient reversal reproduced the alternating-update results almost exactly, confirming the two training procedures are equivalent rather than one being a weaker implementation of the idea. Neither beat the adaptive mechanism's own frontier point; both results support that the frontier itself, not the specific training procedure, is the binding constraint.

7 Limitations and Ethical Considerations

Binary sensitive attributes. Race in COMPAS (restricted to African-American vs. Caucasian) and gender in German Credit are both treated as binary – an artifact of how these datasets record the attribute, not a design choice this method endorses. Neither dataset tests intersectional or non-binary protected groups.

Narrow fairness definitions. DEO and DAO are two specific equalized-odds-family definitions among many possible ones; they do not cover calibration parity, individual fairness, or counterfactual fairness.

The accuracy floor is a partially engineered guarantee. The hard clip preventing alpha from increasing while accuracy is below a floor means "accuracy is preserved" is, to that extent, designed in rather than a purely emergent property of the dual-ascent mechanism. This should be read as a disclosed design choice, not a surprising empirical discovery.

The controller's own hyperparameters were tuned. η (the ascent step size) and the 30% fraction used to set $\epsilon_{\text{DEO}}, \epsilon_{\text{DAO}}$ were chosen by the authors based on what produced a working mechanism within a 30-epoch budget, not derived from first principles or their own independent search. This is the same kind of tuning effort that produced the fixed-alpha values being compared against; a reviewer should weigh the comparison accordingly.

The post-processing finding is conditional, not fully characterized. Two datasets are enough to show that post-processing's COMPAS-win does not generalize unconditionally, and that calibration-set size is the likely mechanism – but not enough to establish the precise data-size threshold at which post-processing stops being reliable, or whether other factors (class balance, group-size imbalance, dimension of the feature space) also contribute. This is flagged as an open question rather than a settled one.

Held-out evaluation feeds the controller's own decisions. For the adaptive method, DEO/DAO are computed on the test split every epoch specifically to drive alpha's update – standard for a train-time controller, but worth stating explicitly: the test split is never used for gradient updates, only for the controller's feedback signal and final reporting.

8 Conclusion

Adaptive Alpha Control removes the need to hand-sweep a fixed weight in adversarial debiasing, reaching a comparable operating point automatically via dual ascent on two explicit fairness constraints. But the more consequential finding in this paper is that a much simpler post-hoc calibration method outperforms it – conditionally. That condition (calibration-set size) is itself the useful takeaway: the choice between in-processing and post-processing debiasing should be informed by how much held-out data is actually available, not by a single-dataset benchmark result in either direction. Future work could characterize this threshold more precisely, test whether a

hybrid (post-processing when data allows, adversarial training as a fallback when it doesn't) outperforms either alone, and extend the two-constraint dual-ascent idea to non-binary or intersectional sensitive attributes.

References

Agarwal, A., Beygelzimer, A., Dudík, M., Langford, J., & Wallach, H. (2018). A Reductions Approach to Fair Classification. In *Proceedings of the 35th International Conference on Machine Learning (ICML)*, PMLR vol. 80, pp. 60–69.

Angwin, J., Larson, J., Mattu, S., & Kirchner, L. (2016, May 23). Machine Bias. *ProPublica*.

Cotter, A., Jiang, H., Gupta, M., Wang, S., Narayan, T., You, S., & Sridharan, K. (2019). Optimization with Non-Differentiable Constraints with Applications to Fairness, Recall, Churn, and Other Goals. *Journal of Machine Learning Research*, 20(172), 1–59.

Ganin, Y., & Lempitsky, V. (2015). Unsupervised Domain Adaptation by Backpropagation. In *Proceedings of the 32nd International Conference on Machine Learning (ICML)*, PMLR vol. 37, pp. 1180–1189.

Hardt, M., Price, E., & Srebro, N. (2016). Equality of Opportunity in Supervised Learning. In *Advances in Neural Information Processing Systems 29 (NeurIPS)*, pp. 3315–3323.

Zhang, B. H., Lemoine, B., & Mitchell, M. (2018). Mitigating Unwanted Biases with Adversarial Learning. In *Proceedings of the 2018 AAAI/ACM Conference on AI, Ethics, and Society (AIES)*.